Poster 1 — Who wants to be rich?

DOK 3

Data

Super Status! 433 high school students answered two questions: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

The student newsletter wrote: "Freeze Time people are the ones who want to be rich. 26 of them chose Rich, more than any other superpower."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. For each superpower, what percent of its fans chose Rich? Write a fraction and a percent. Which total is the denominator, and why?

b. Draw a segmented bar chart: one 100% bar per superpower, split into Famous / Happy / Healthy / Rich. Whole percents are fine. Add a key and a percent axis.

c. Verdict. Do Freeze Time fans stand out for wanting to be rich? Compare their percent with the other four groups and with all 433 students.

d. Repair. Rewrite the claim so it is true. Use a percent, not a count.

e. Is there an association between superpower and status? Compare two bars that look different (try Telepathy and Fly) and explain in one sentence.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 1 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ Five "Rich" percents, each with its fraction
- ☐ Segmented bar chart: five bars, each 100%, four segments, a key, a percent axis
- ☐ A verdict that compares Freeze Time's percent with the other groups and with all 433 students
- ☐ The repaired claim, true, with a percent
- ☐ One association sentence comparing two superpowers' bars

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The five Rich fractions and percents, then the percents for the bars.	The segmented bar chart: five 100% bars, four segments, key, percent axis, title.	The verdict, the repaired claim, the association sentence.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: given · denominator · association

Sketch the segmented bar chart here first (five bars, percent axis, key).

Draft the poster sentences here. Every sentence names the context and uses a number from the table.

Poster 2 — The flipped headline

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

A headline in the school paper: "Most students who want to be Famous chose Fly. So most students who chose Fly want to be Famous."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Find P(Fly | Famous) and P(Famous | Fly) as fractions and percents. Which total is the denominator each time, and why?

b. Shade the "given" group on two copies of the table, one for each probability. Circle the cell the two share.

c. Verdict on each half. Is "most" right in the first sentence? Does the second sentence follow from the first?

d. Repair. Rewrite the headline as two true sentences, each with its probability.

e. Build. Find another pair in the table where flipping the "given" changes the answer a lot. Write both probabilities and explain in one sentence why they differ.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 2 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ Both conditional probabilities as fractions and percents
- ☐ Two copies of the table, each with its "given" group shaded
- ☐ A verdict on each half of the headline, each with its number
- ☐ The repaired headline: two true sentences with their numbers
- ☐ A second flipped pair from the table with both probabilities and a one-sentence why

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The two fractions and percents, and the second flipped pair. Name each denominator.	The two shaded tables, with the shared cell circled.	The verdict on each half, the repaired headline, the why sentence.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: given · denominator

Sketch the two shaded tables here first.

Draft the poster sentences here.

Poster 3 — Can't happen together, so unrelated?

DOK 3

Data

Super Status! 433 high school students answered: "What superpower would you choose?" and "What status would you prefer?" One of these 433 students is picked at random.

Superpower	Famous	Happy	Healthy	Rich	Total
Fly	5	48	14	22	**89**
Freeze Time	3	63	9	26	**101**
Invisibility	3	62	11	22	**98**
Super Strength	4	43	17	13	**77**
Telepathy	0	49	0	19	**68**
Total	**15**	**265**	**51**	**102**	**433**

The claim you are judging

"Nobody who chose Telepathy wants to be Famous: 0 of 68. Since the two never happen together, Telepathy and Famous are independent. One has nothing to do with the other."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Are "chose Telepathy" and "wants to be Famous" mutually exclusive? Show the joint probability P(Telepathy and Famous).

b. Are they independent? Compare P(Famous) with P(Famous | Telepathy).

c. Verdict, in one sentence that uses both terms correctly. What does knowing "Telepathy" do to the chance of "Famous"?

d. Prove it in general. Two events can each happen, and they are mutually exclusive. Explain why they can never be independent. (What does independence say P(A and B) must equal?)

e. Build. Make a 2x2 table of 100 students for "plays a sport" and "plays an instrument" where the two events are independent but NOT mutually exclusive. Write the check under it.

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 3 — checklist, roles, sketch

Can't happen together, so unrelated?

Poster checklist (this is the rubric)

- ☐ The joint probability P(Telepathy and Famous) from the table, and what it tells you
- ☐ The independence test, with its two numbers
- ☐ One verdict sentence that uses *mutually exclusive* AND *independent* correctly
- ☐ The proof from part (d), in two or three sentences
- ☐ Your 2x2 table (100 students) with the independence check written under it

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The probabilities, and the numbers in your 2x2 table.	The big table with the Telepathy row and the Famous column shaded, and your 2x2 table with its check underneath.	The verdict sentence and the proof.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: mutually exclusive | independent | given

Sketch your 2x2 table here first.

Draft the poster sentences here.

Poster 4 — Double counted

DOK 3

Data

A survey of 2,500 adults asked about education level and where they get their news. One of these 2,500 adults is picked at random.

Group	Count
College graduates	693
Get news mainly from the internet	687
College graduates who get news from the internet	245
All adults surveyed	**2,500**

The claim you are judging

A blog wrote: "P(college grad or internet news) = 693/2500 + 687/2500 = 55.2%. And college grads are no more likely than anyone else to get their news online: only 245/2500 = 9.8% of adults are grads who read news online."

Two mistakes are hiding in this claim. Find both.

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Draw a Venn diagram (or a two-way table) and fill in all four regions: both, grad only, internet only, neither. Check that they add to 2,500.

b. Find the mistake in the 55.2%. Compute the correct P(grad or internet). Which people were counted twice?

c. The blog backs its second sentence with 9.8%. Is 9.8% a joint or a conditional probability? Which probability does "no more likely than anyone else" really depend on?

d. Test the second sentence with the right probability. Are the two events independent? Who is more likely to read news online?

e. Build. Keep 693 college grads and 687 internet readers. How many people in "both" would independence predict? Compare that with 245. What does the gap tell you?

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 4 — checklist, roles, sketch

Double counted

Poster checklist (this is the rubric)

- ☐ Four regions filled and summing to 2,500 (both / grad only / internet only / neither)
- ☐ The people counted twice, named, and the corrected union as a fraction and a percent
- ☐ The blog's 9.8% named (joint or conditional?) and the probability it should have used
- ☐ The independence test with its two numbers, and a verdict naming who is more likely
- ☐ The "both" count independence would predict, compared with 245

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The four regions, the union, the test, the predicted "both" count. Show every subtraction.	The Venn diagram with the overlap shaded (or the 2x2 table), every region labeled with its count.	The double-count sentence, the sentence about the 9.8%, the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: union · given · independent

Sketch the Venn diagram here first (two circles, four regions, counts).

Draft the poster sentences here.

Poster 5 — Is the game fair?

DOK 3

Data

A carnival game costs $2 to play. You roll a fair six-sided die. Roll a 6: you win $10. Roll a 1: you win $3. Otherwise: you win nothing.

The claim you are judging

The sign on the booth: "A FAIR GAME! You win on 2 of every 6 rolls."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Let X = the player's net profit for one game (winnings minus the $2). List every value of X with its probability in a table. Check that the probabilities add to 1.

b. Compute the mean μ_X and the standard deviation σ_X. Show the formula and the substitution; a calculator may do the arithmetic.

c. Interpret μ_X and σ_X in context. Use "in the long run" and say "per game".

d. Verdict. Is the game fair? Who does it favor? Why is "win on 2 of every 6 rolls" the wrong number for judging fairness?

e. Repair. Change ONE payout so that $\mu_X = 0$ exactly. Show the new table and the new mean.

f. Simulate 20 games. On the TI-84: randInt(1,6,20) STO→ L1, then read the 20 rolls in STAT → EDIT. Use your table to turn each roll into its net, and find the average. Why does it differ from μ_X? What would 1,000 games do?

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 5 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ The distribution table of X = net profit, with probabilities that sum to 1
- ☐ μ_X and σ_X, each with its formula and substitution
- ☐ Both interpretations in context, with "in the long run" and "per game"
- ☐ A verdict naming who the game favors and why "2 of 6" is the wrong number
- ☐ The one-payout fix (new mean shown), and a dotplot of your 20 simulated games with their average marked

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
The table, μ, σ, the fix. Show the formula before the numbers.	Runs the 20 games on the TI-84. The dotplot of the 20 nets, with their average and μ_X both marked.	The two interpretations and the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____." **Also:** "In the long run, _____ per _____."

Your words: expected value · long run · net

Sketch the dotplot of your 20 simulated games here first. (randInt is under MATH → PROB.)

Draft the poster sentences here.

Poster 6 (spare) — Rigged coin?

DOK 3

Data

You make the data: 20 flips of a fair coin on the TI-84, where 1 = heads. randInt(0,1,20) STO→ L1, then read the flips down L1 in STAT → EDIT. (randInt is under MATH → PROB.)

The claim you are judging

"I flipped a coin 20 times and got 6 heads in a row. That can't happen with a fair coin. It's rigged."

Solve together first. Scratch paper and whiteboards, no poster yet.

a. Design. Write the simulation in three sentences: how one flip is modeled, what one trial is, and what you record from each trial (the longest run of heads in the 20 flips).

b. Run 25 trials. Record the longest run each time. Draw a dotplot of the 25 values.

c. Estimate P(longest run is 6 or more) from your dotplot.

d. Verdict. First decide: how rare would 6 in a row have to be before you believed the coin was rigged? Then compare your estimate with that bar. Is 6 in a row convincing evidence?

e. Your estimate is not the exact answer. Why not? What would 1,000 trials do? (Your teacher has the exact value.)

Our verdict in one sentence (show your teacher before it goes on the poster):

Poster 6 — checklist, roles, sketch

Poster checklist (this is the rubric)

- ☐ The design in three sentences (flip, trial, what is recorded)
- ☐ 25 trials recorded in a table
- ☐ A dotplot of the 25 longest runs with a scale and a title
- ☐ The estimated probability as a fraction of 25 and a decimal
- ☐ The verdict, plus one sentence on what more trials would do

All five = E · three or four = P · fewer = I

Table Keeper	Visual	Writer
Runs the calculator, records the 25 longest runs, computes the fraction.	The dotplot, with a scale that covers every value you got and the "6 or more" region shaded.	The design sentences and the verdict.

Verdict frame: "The claim says _____. The number it really depends on is _____ = _____. So the claim is (supported / partly supported / not supported) because _____."

Your words: simulation · trial · estimate

Record the 25 trials and sketch the dotplot here first.

Draft the poster sentences here.